

Enterprise Reporting & Automation Strategy Using Microsoft Tools

Company : RetailHub Pvt Ltd

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Proposed Solution Overview

RetailHub currently relies on manual spreadsheet-based reporting, which leads to delays, data inconsistencies, and limited visibility into business operations. To modernize its reporting and automation capabilities, the organization can implement a solution using Microsoft Power Platform and SQL Server Reporting Services (SSRS).

The proposed solution integrates data from multiple systems such as sales transactions, inventory systems, customer orders, supplier databases, and finance systems into a centralized SQL Server database. This centralized data repository will act as the foundation for reporting and analytics.

SSRS will be used to generate structured operational reports such as financial statements, daily sales reports, and invoice reports. These reports require fixed formatting and scheduled delivery.

At the same time, Power BI will provide interactive dashboards that allow management to analyze sales trends, regional performance, and inventory levels in real time.

To improve operational efficiency, Power Apps can be used to create internal business applications for inventory tracking and operational management, while Power Automate will automate workflows such as purchase approvals and report notifications.

Additionally, Power Virtual Agents can be used to deploy a chatbot that helps customers track orders and receive automated responses to common queries.

Together, these Microsoft tools create an integrated ecosystem that enhances reporting capabilities, improves decision-making, reduces manual work, and enhances customer experience.

User Story 1 – Understanding the Current Reporting Challenges

As a Technology Consultant, I want to analyze the existing reporting process so that I can identify the major operational and data-related challenges faced by RetailHub.

Currently, RetailHub relies heavily on spreadsheets for reporting across different departments. Store managers receive sales reports manually, finance teams prepare monthly financial summaries through manual calculations, and inventory managers lack real-time visibility of stock levels.

These processes lead to several operational challenges:

- Manual reporting increases the risk of data errors and inconsistencies.
- Decision-making becomes slow due to delayed reporting.
- Employees spend excessive time collecting and compiling data from multiple systems.
- There is limited visibility into business performance across regions.
- Customer support teams must handle repetitive queries manually.

Replacing spreadsheet-based reporting with automated reporting tools can significantly improve operational efficiency.

User Story 2 – Business Intelligence for Management

As a Store Manager or Business Executive, I want to access interactive dashboards showing sales performance and business trends so that I can make data-driven decisions quickly.

Solution Using Power BI

Power BI dashboards can provide management with insights such as:

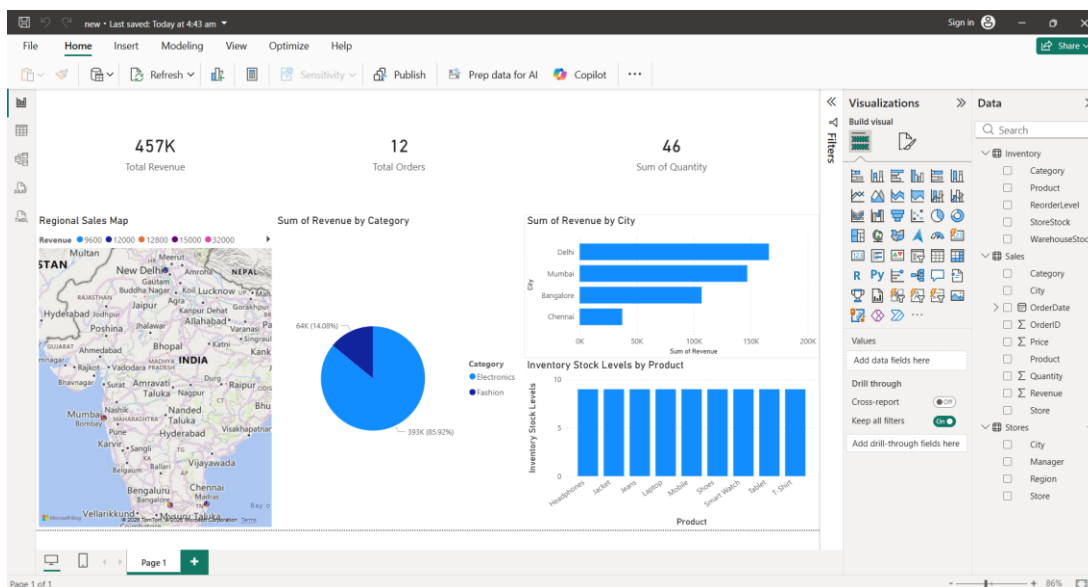
- Sales performance by store and region
- Monthly and quarterly revenue trends
- Product category performance
- Customer purchasing patterns
- Inventory availability across stores

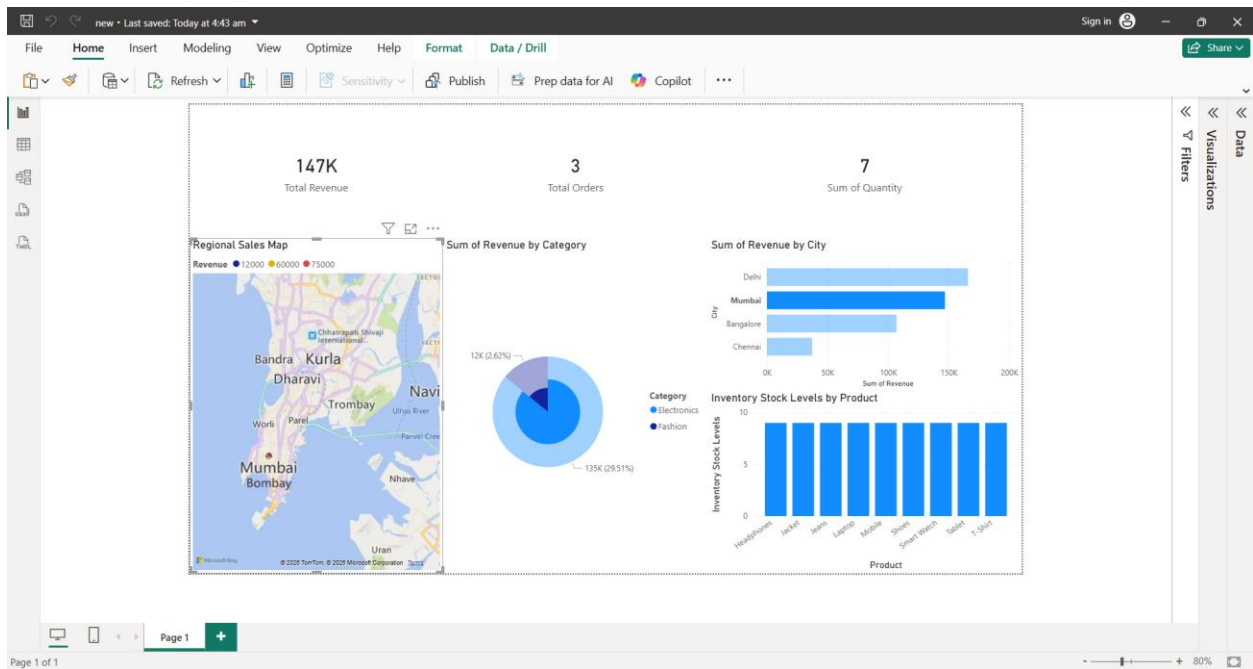
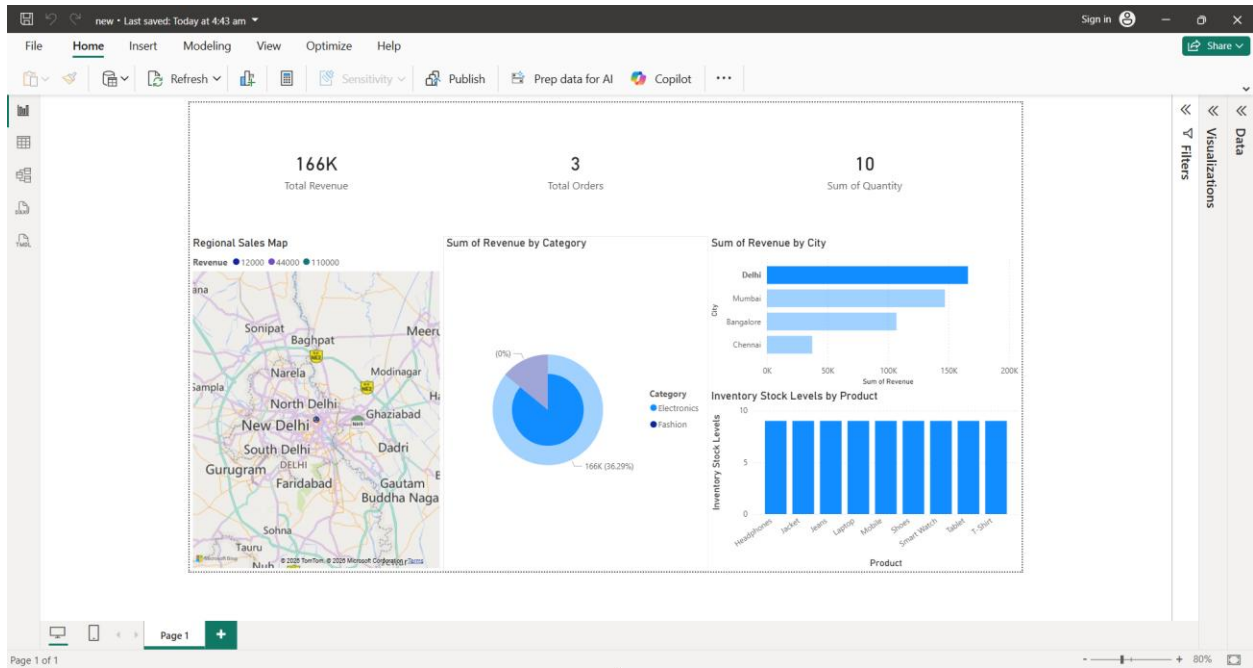
Managers can interact with dashboards by applying filters, drilling down into specific regions, and identifying trends that influence business strategy.

This enables faster and more informed decision-making.

Implementation Demonstration

To demonstrate the proposed business intelligence solution, a Power BI dashboard was developed by connecting Power BI Desktop to the RetailHub SQL Server database. The dashboard provides interactive visualizations such as KPI cards, bar charts, pie charts, and geographic maps to analyze sales performance across different cities and product categories.





User Story 3 – Improving Internal Business Applications

As an Inventory Manager, I want a centralized application to manage inventory data so that I can monitor stock levels across stores and warehouses.

Solution Using Power Apps

Power Apps can be used to build an internal inventory management application that allows employees to:

- Update product stock levels
- Track inventory movement between warehouses and stores
- Manage supplier orders
- Monitor stock shortages

This application can be accessed through web browsers or mobile devices, allowing staff to update inventory information in real time.

User Story 4 – Automating Business Workflows

As an Operations Manager, I want repetitive operational tasks to be automated so that employees can focus on more strategic work.

Solution Using Power Automate

Power Automate can automate various workflows including:

- Approval workflows for purchase orders
- Automatic notifications for low inventory levels
- Scheduled report distribution
- Alerts for delayed shipments

Automation reduces manual work and improves operational efficiency.

User Story 5 – Enhancing Customer Support

As a Customer Support Manager, I want customers to receive automated responses for common queries so that support teams can focus on more complex issues.

Solution Using Power Virtual Agents

A chatbot can be implemented to assist customers with:

- Checking order status
- Tracking delivery updates
- Answering frequently asked questions

This reduces the workload of customer support agents and improves response time for customers.

User Story 6 – Generating Operational Reports

As a **Finance Manager**, I want standardized operational reports to monitor financial and operational performance.

Solution Using SSRS

SSRS will generate structured reports such as:

- Daily store sales reports
- Monthly financial statements
- Inventory summary reports
- Customer invoices and billing reports

These reports can be automatically scheduled and distributed to relevant stakeholders in formats such as **PDF or Excel**.

SQLQuery3.s...TUSHA (52)

```
1 SELECT * FROM inventory;
2 SELECT * FROM Sales;
3 SELECT * FROM Stores;
```

100 % No issues found

Product	Category	WarehouseStock	StoreStock	ReorderLevel
Laptop	Electronics	50	20	15
Mobile	Electronics	120	60	30
Tablet	Electronics	40	18	10
Smart Watch	Electronics	70	30	20
Headphones	Electronics	150	80	40
Shoes	Fashion	200	90	50
T-Shirt	Fashion	300	150	80
Jeans	Fashion	180	70	40

OrderID	OrderDate	Store	City	Product	Category	Quantity	Price	Revenue
1001	2025-01-01	Store A	Delhi	Laptop	Electronics	2	55000	110000
1002	2025-01-02	Store B	Mumbai	Mobile	Electronics	3	25000	75000
1003	2025-01-03	Store C	Bangalore	Shoes	Fashion	5	3000	15000
1004	2025-01-03	Store A	Delhi	Head...	Electronics	6	2000	12000
1005	2025-01-04	Store D	Chennai	T-Shirt	Fashion	8	1200	9600
1006	2025-01-05	Store B	Mumbai	Laptop	Electronics	1	60000	60000
1007	2025-01-06	Store C	Bangalore	Smart...	Electronics	4	8000	32000
1008	2025-01-06	Store D	Chennai	Jeans	Fashion	6	2500	15000

Store	City	Region	Manager
Store A	Delhi	North	Rahul Sharma
Store B	Mumbai	West	Amil Patel
Store C	Bangalore	South	Priya Nair
Store D	Chennai	South	Arun Kumar

Query executed successfully.

(localdb)\MSSQLLocalDB (11... TUSHAR449\TUSHA (52) RetailHubDB 00:00:00 Row: 1, Col: 1 25 rows

User Story 7 – Selecting the Appropriate Reporting Tool

As a Business Analyst, I want to understand when to use SSRS versus Power BI so that reporting tools are used effectively.

Criteria	SSRS	Power BI
Reporting Type	Paginated reports	Interactive dashboards
Best Use Case	Financial and operational reports	Business analytics
Visualization	Basic charts and tables	Advanced visualizations
User Interaction	Limited	Highly interactive
Output	PDF, Excel, Print	Web dashboards

SSRS should be used for operational reporting, while Power BI is best suited for analytics and performance monitoring.

User Story 8 – Designing the System Architecture

As an IT Architect, I want to design a unified architecture integrating Microsoft reporting and automation tools so that data flows efficiently across the organization.

Architecture Flow

